QPCode **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) MODEL EXAMINATION October 2026

Fifth Semester

SIGNATURE COURSE - MG5SECPHYA00 - RESEARCH METHODOLOGY AND SCIENTIFIC TYPESETTING

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ with options

Answer all questions

Each question carries 1 mark

1. Which research approach primarily develops or tests ideas through mathematical or conceptual reasoning? [Understand] / [CO1]

A. Use of concepts, models and logical reasoning
B. Computational research
C. Theoretical research
D. Computational research

2. Basic research is primarily undertaken to [Understand] / [CO1]

A. Develop or test knowledge, theories and hypotheses
B. Basic research
C. Advancement of knowledge
D. Solve problems of immediate practical concern

3. What is the main role of computational research? [Remember] / [CO1]

A. Computational tool
B. Computational research
C. Algorithms and numerical calculations
D. Using computation to model, simulate or analyse problems

4. Research problems are often found by [Understand] / [CO1]

A. Reading literature and identifying gaps or unresolved issues
B. Generate new or unexpected findings
C. Lead to meaningful discoveries or understanding
D. Formally test or challenge the finding

5. The first step in identifying a research problem is often to [Remember] / [CO1]

A. Recognize an issue, gap or unanswered question
B. The chosen field of study
C. Clearly defined and researchable
D. Something important is not adequately known

6. Defining a research problem means [Remember] / [CO1]

A. Unclear and overly broad wording
B. Variables, context or population where appropriate
C. Specifying exactly what will be investigated
D. Delimiting the study population/context

7. A research question is primarily intended to [Understand] / [CO1]

A. Guide what the study seeks to answer
B. Clearly formulated research questions
C. A research question

D. How does temperature affect the resistance of a given material?

8. A hypothesis is best described as [Remember] / [CO1]
A. A theory
B. Testable
C. A scientifically reasonable prediction
D. Reconsidered or revised

9. Research design is best described as [Understand] / [CO1]
A. A factor that can take different values
B. The experimental condition being tested
C. The purpose of each step
D. The overall plan for conducting the study

10. Accuracy refers to [Remember] / [CO1]
A. Unreliable or biased results
B. Closeness of a measurement to the true or accepted value
C. Bias experimental conclusions
D. Recorded carefully

11. Quantitative data are data that [Understand] / [CO2]
A. Can be represented numerically
B. Secondary data
C. Qualitative data
D. Quantitative data

12. The main purpose of a literature review is to [Remember] / [CO2]
A. Research gaps
B. Describe and evaluate literature relevant to a topic
C. An overview of the field of inquiry
D. The research problem or question

13. Which is an effective literature-search strategy? [Understand] / [CO2]
A. Digital academic information
B. Use focused keywords related to the research question
C. Refining terms based on relevant results
D. Indian theses and dissertations

14. A literature review should contain [Remember] / [CO2]
A. The basis for selecting the literature
B. Introduction, body and conclusion
C. Major agreements, disagreements and general conclusions
D. Principal questions asked in the field

15. The final year report begins with [Understand] / [CO2]
A. The issue that needs investigation
B. What new value the research provides
C. The specific tasks needed to achieve the project goals
D. An abstract

16. The methodology section primarily explains [Remember] / [CO2]
A. A project proposal
B. How the research was conducted
C. Project schedule and milestones
D. Resources and time needed for the work

17. The Results section should primarily [Understand] / [CO2]
A. Present the findings of the study
B. Interpret findings in relation to the research question
C. Observations, data and the hypothesis
D. Words, tables and graphics

18. Interpretation of results asks [Remember] / [CO2]

A. Bring together the main findings
B. What the findings mean in relation to the research question
C. The research question and evidence
D. Illustrate relevant findings

19. Research ethics concerns [Understand] / [CO2]
A. Responsible conduct of research
B. Honest reporting of data and methods
C. The integrity and credibility of research
D. Submitting the same research work to more than one journal

20. Plagiarism involves [Remember] / [CO2]
A. Restating meaning in your own words
B. Using another person's work or ideas without proper credit
C. An accurate reference
D. Identified as a quotation and cited

21. A direct quotation is [Understand] / [CO3]
A. Understanding the original text
B. Where the borrowed information originated
C. An exact reproduction of spoken or written words
D. Sparingly and for a clear purpose

22. LaTeX is primarily a [Remember] / [CO3]
A. High-quality typesetting
B. Document preparation system
C. Professional typesetting and mathematical support
D. Automatic numbering and referencing

23. A LaTeX document has two main parts: [Understand] / [CO3]
A. \documentclass
B. Articles and reports
C. Letters
D. Preamble and document body

24. The command for bold text is [Remember] / [CO3]
A. \texttt{}
B. enumerate
C. \textit{}
D. \textbf{}

25. The environment used to create tables is [Understand] / [CO3]
A. \hline
B. Columns
C. tabular
D. Laboratory observations

26. The package required for including graphics is [Remember] / [CO4]
A. \caption{}
B. graphicx
C. \centering
D. Describe the figure clearly

27. Automatic cross-referencing is useful for [Analyse] / [CO4]
A. Sections, figures, tables and equations
B. Point to the intended element
C. Renumber items manually
D. Maintains consistency when content changes

28. BibTeX is mainly used for [Analyse] / [CO4]
A. Sources cited or referenced in a document
B. Scientific writing
C. Bibliography management

D. Managing bibliographic entries

29. Which sequence best represents the essential structure of a scientific project report prepared using LaTeX? [Apply] / [CO4]

A. Title/abstract → introduction → methodology → results and discussion → conclusion/future work → references
B. Title → CV → acknowledgements only
C. Figures → bibliography → title only
D. Tables → equations → references only

30. Which combination of LaTeX features is most useful for professional scientific documentation and CV preparation? [Apply] / [CO4]

A. Structured sections, tables, figures, equations, lists and consistent formatting
B. Only manual page formatting
C. Only image editing
D. Only plain text without document structure

Part B

Essay Question

Answer the following question

Each question carries 20 marks

31. Prepare a scientific project report titled “Experimental Investigation of the Dependence of Resistance on Temperature” using LaTeX. Explain and demonstrate how you would prepare the report containing: (a) title page and abstract, (b) introduction and objectives, (c) literature review, (d) experimental methodology, (e) results and discussion, (f) tables containing experimental observations, (g) figures/graphs representing experimental results, (h) conclusion and future scope, and (i) references. Write suitable LaTeX commands/environments for creating the major elements of the report.

[Apply] / [CO3 / CO4]

***** End of Question Paper *****